

LEBANESE
INTERNATIONAL
UNIVERSITY



Lebanese International University
School of Arts and Sciences
Department of Mathematics & Physics

Summer 2023-2024

Math 225 – Linear Algebra with Applications (3 Cr.)

Instructor:
E-Mail:
Campus: Beirut

Office:
Office Hours: | |
Lectures time: | |

A. Course Description

This course gives an introductory treatment of Linear Algebra and it is suitable as a first undergraduate course.

B. Course Objectives

This course aims to:

1. Make the students knowledgeable about the basics of Linear Algebra.
2. Help the students in understanding the link between matrices and solving linear systems.
3. Provide the students with the concepts of vector spaces and inner product spaces.

C. Student's Learning Outcomes

Upon successful completion of this course, the student should be able to:

1. Solve systems of linear equations using matrices and their properties, including inverses and determinants.
2. Understand and apply vector space concepts and operations, such as linear combinations, span, linear independence, basis and dimension.
3. Compute inner product spaces concepts and operations, including orthogonality and orthonormality.
4. Apply matrix diagonalization in simplifying computations.

D. Topics Covered and detailed schedule

Lecture	Topic	Examples & Exercises
1	2.1 Operations with matrices. 2.2 Properties of matrix operations	2.1 : 1,2,3,4,5 2.2 : 2,4,5, 6.
2	2.2 Properties of matrix operations 1.1 Introduction to systems of Linear Equations	2.2 : 7,8,9,10 1.1 : 1,2,3
3	1.2 Gaussian Elimination and Gauss-Jordan Elimination	1.2 : 1,4,5,9
4	2.3 The inverse of a matrix	2.3 : 1,2,3,4,5,6,7,8
5	Solving session	
6	3.1 The Determinant of a matrix 3.2 Evaluation of a Determinant Using Elementary Operations	3.1: 1,2,3,4,6 3.2 : 2,3,4,5,6
7	3.3 Properties of Determinants 3.5 Applications of Determinants	3.3 : 1,2,3,4,5,6 3.5 : 1,2,3,4
8	Exam 1 (Tentative)	
9	4.1 Vectors in R^n 4.2 Vector Spaces 4.3 Subspaces of Vector Spaces	4.1: 4, 5, 6 4.2 : 1,2,3,4,6,7,8 4.3 : 2,3
10	4.3 Subspaces of Vector Spaces 4.4 Spanning sets and Linear Independence	4.3 : 4,8 4.4:2,3,4,5,6,7,8,9,10,11
11	4.5 Basis and Dimension Solving session	4.5 : 1,2,4,5,6,7,8,9,11,12
12	4.6 Rank of a matrix and Systems of Linear Equations	4.6 : 1,2,3,4,5,6,7,8
13	Exam 2 (Tentative)	

Lecture	Topic	Examples & Exercises
14	7.1 Eigenvalues and Eigenvectors	7.1 : 1,2,4,7,8
15	7.2 Diagonalization Solving session	7.2 : 1,2,3,4,5,7
16	5.1 Length and Dot product in R^n 5.3 Orthonormal Bases : Gram-Schmidt Process	5.1 : 1,2,3,4,5,6,7,8,9,10 5.3: 1, 4, 6, 7, 8
17	5.2 Inner product spaces 6.1: Introduction to Linear Transformations	5.2: 2, 3, 4, 6 6.1: 1, 2, 3, 4, 5, 6, 11
18	Revision	

E. Textbook

R. Larson and D. C. Falvo, *Elementary linear algebra with applications*, Houghton Mifflin Harcourt Publishing, 6th edition.

F. Prerequisite – Corequisite

Corequisites: Math210

G. Course Assessment

Evaluation	Weight	Date and time	Chapters Covered	LO
Exam 1	27%		Chapters 1 and 2.	LO 1
Exam 2	27%		Chapters 3 and 4.(1,2,3,4,5)	LO 1: 60% LO 2: 40%
Attendance	6%			
Final	40%		Chapters 4, 5, and 7	LO 2: 35% LO 3: 35% LO 4: 30%

Mapping SLOs with PLOs:

Student's learning Outcomes	Program Outcomes							(8)	(9)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)		
SLO1		X				X			
SLO2		X							
SLO3		X							
SLO4		X							